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CredEx Fintech: Business Model Transformation During the Digital Era

On the last day of 2017, Tang Xia, Chairman of CredEx Fintech, attended a special event to celebrate the coming New Year. Specifically, Wu Xiaobo, one of the most prestigious and bestselling authors of China's business history, was presenting his end-of-year talk show titled "Forecast 2018" at Ling Shan Brahma Place in Wuxi, China, and had invited Tang as one of his VIP guests. In the talk show, Wu reviewed the spectacular 40-year journey of China's economic reform and summarized the driving factors of social progress as two forces: institutional change and technological change. This observation struck a resonant chord with Tang. CredEx's path of growth since its inception and its current challenges both bore the marks of these two forces.

In 2017, these two intertwining forces pushed CredEx to an intersection of choices. Since its founding in 2010, CredEx had been at the forefront of financial technology (fintech) innovation in the microfinance market. In just a few years, the loan balance under its management increased from around RMB 400 million^a to more than RMB 5 billion. In particular, after CredEx launched its mobile app in 2015, its loan balance under management had quintupled within one year. At the same time, CredEx had forged a drastic digital transformation of its business model. As a result, its labor force was reduced from more than 3,000 people at its peak to around 300 people, and both the accuracy and efficiency of its risk management significantly improved. Moreover, the company's one-stop microfinance solution had received numerous innovative awards and attracted attention from industry experts and research institutions.

Nevertheless, these achievements did not make Tang feel more relaxed as he sensed that profound changes were taking place in the external environment of the microfinance business. With the increased adoption of mobile as the major option for customer access in the microfinance industry, CredEx's advantage in its mobile-based customer acquisition had become less significant. At the same time, the rapid development of the microfinance industry had resulted in an increase in both the leverage ratio and the default risk of microfinance customers. A more alarming factor was that China's economic growth had started slowing down, which would magnify the risk of microfinance from various aspects.

^a 1 RMB = 0.14 USD, <https://www1.oanda.com/currency/converter/>, accessed December 2019.

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However, the rising risk did not prevent competition in the industry from intensifying – starting in 2014, multiple internet giants with strong technological capabilities had also entered the microlending and consumer finance fields.

What made Tang even more concerned was that China's regulators had been substantially changing their attitudes toward the microlending industry due to the social unrest and chaos caused by an increasing number of material default events in the microlending and peer-to-peer (P2P) lending sectors. Prior to 2015, the government had basically adopted policies that primarily facilitated and supported microlending business. However, since 2016, it had frequently launched specific regulatory requirements for the industry and initiated rigid "cleanup and rectification" measures against noncompliant businesses. On December 1, 2017, the People's Bank of China and the China Banking Regulatory Commission jointly issued the *Notice on Regulating and Rectifying "Cash Loan" Business*, referred to as "Circular 141" within the industry, imposing more stringent restrictions on cash loans, online microloans, and the ways in which banks and other financial institutions could get involved in those businesses. It was the practitioners' common view that Circular 141 tremendously impacted the potential growth of microlenders and challenged the legitimacy of their business models, marking an essential change of the government's attitude toward regulating the microfinance industry.

All these changes led Tang to review and reflect on CredEx's business model and future. As Wu finished his talk show, Tang posted an excerpt of Wu's talk on his WeChat Moments:^b "This era has descended with irresistible force. All things spring into life without restraint, dust frisks along with the first light of morning. Brooks and creeks converge into rivers. Nameless hills rise as peaks. This is the time when the world embraces a vast openness."

Tang felt that Wu's words painted a portrait of China's microfinance industry in the past few years. Against this backdrop, CredEx had achieved its high-speed development. However, the intertwining technological and regulatory changes in the financial industry presented new opportunities and challenges to CredEx. As a fintech pioneer, how should CredEx innovate its business model to maintain its sustained competitive advantages in the highly volatile microfinance industry? This was the tough question that Tang and the CredEx management team had to face and answer.

The Microfinance Industry in China

Microloans were small loans made to individuals or small and medium-sized enterprises (SMEs), providing funds for personal consumption, SME working capital, or SME business expansion. Most microloans ranged from a few thousand RMB to tens of thousands of RMB. The pioneering microlending practices could be traced back to the Grameen Bank of Bangladesh, which provided loans to villages that did not have access to commercial loans and built a risk management system through social trust relationships. The project had achieved an impressive performance of high yield (20% simple interest) and low risk (a loan recovery rate of 98%).¹ Its founder, Muhammad Yunus, was awarded the Nobel Peace Prize in 2006.

China tentatively began microlending operations in 1993 with an initial focus of alleviating poverty in rural areas. The economic growth in China stimulated a surging demand for funding for the business development of micro and small enterprises (MSEs) as well as personal consumption. According to China's Ministry of Industry and Information Technology (MIIT), MSEs accounted for 97.3% of the country's total enterprises in 2011.² The final consumption expenditure increased from RMB 13.6

^b WeChat Moments was a social-networking function of the smartphone app WeChat through which users could share and get access to their friends' information and create private communication circles.

trillion in 2007 to RMB 19.9 trillion in 2010. (See **Exhibit 1** for detailed data on China's economic and consumer credit growth.) However, due to multiple reasons involving cost and risk, China's financial system, especially the banks, could not effectively meet these new needs for microfunding, which accelerated the emergence of commercial microlending companies. In 2005, Chinese regulators started issuing licenses to private microlenders for the first time, together with rectification and supervision measures. In 2008, the power of license approval was delegated to the provincial government, stimulating the development of the microlending industry through deregulation. Such measures led to the explosive growth of microlending enterprises in the following few years. In 2008, there were fewer than 500 microlending companies registered in China;³ by 2015, this number approached 9,000. Total loan balance increased from less than RMB 200 billion in 2010 to nearly 1 trillion RMB in 2015. (See **Exhibit 2** for detailed data on the growth of China's microfinance industry.)

From the perspective of traditional financial institutions (FIs), microlending was associated with features such as high yield, high risk, and high operational costs; the two latter features were the critical reasons that deterred traditional FIs from engaging in microlending. Traditional FIs usually adopted the IPC^c risk control model, which originated from the Deutsche PostBank, to handle microloans. At the core of this model was the offline customer acquisition network (e.g., branches), which also played the central role of risk management. Specifically, the IPC model required the client manager at a branch to perform the field investigation, triangulate the information, and exercise judgment about the borrower's ability and willingness to repay the loan. Thus, the effectiveness of risk management inevitably relied on the branch client manager's capability and expertise. In practice, the IPC model inherently invited higher operational risk and moral hazard. The moral hazard was rooted not only in the authenticity of information verified by the client manager's field investigation but also in the conflict of interest between loan risk and sales performance, which would also call into question the accountability of the client managers as they assumed the function of salespeople. In addition, the IPC model relied heavily on human capital, which limited its economies of scale and made its per capita efficiency metrics (such as loan balance per capita and profit per capita) much lower than those of large corporate loans. Therefore, although the return rate of microlending was relatively high,^d traditional FIs did not place a top priority on microlending in their business landscape. They were merely developing their microlending businesses to respond to the regulators' call for inclusive finance.

The absence of traditional FIs in microlending gave the emerging microlenders more space for development. Regulators set clear pricing caps on microloans, and the design of microloans was rather homogeneous. Therefore, microlenders mainly competed in customer experience, customer acquisition efficiency, risk management ability, capital cost, and internal operating efficiency. Specifically, customer experience was mainly indicated in the convenience level for loan applications and the service quality after loan issuance. Customer acquisition efficiency was linked to the speed and cost of acquiring customers in the targeted segments. Risk management ability was measured by the loss rates during the loan collection process. Due to the lagging factor of loan risk, risk management ability could only be sufficiently tested with time and economic cycles. Internal operational efficiency included human resources training, marketing, finance, internal management, and other work that set up the

^c Internationale Projekt Consult (IPC) GmbH was an international consulting company based in Frankfurt am Main, Germany, which was best known for its risk management model in microfinance and SME finance businesses.

^d In 1988, the Supreme People's Court and the Supreme People's Procuratorate issued a judicial explanation on the interest of private lending: "The interest rate of private lending may be appropriately higher than the bank interest rate, but the maximum rate shall not be greater than four times of the bank loan interest rate in the same period. The interest portion in excess of the maximum is not protected by law." In 2015, the Supreme People's Court issued the Regulations of the Supreme People's Court on Several Issues Concerning the Application of Law in the Trial of Private Lending Cases, clearly stipulating that when the interest rate exceeded 36%, the law did not protect the lender's interest portion exceeding 36%.

important foundation for the other capabilities. All of these capabilities determined the cost and pace of the expansion of microlenders and hence were the critical cornerstones for their competitive advantages.

The Early Days of CredEx

In October 2010, Tang Xia and his start-up team founded CredEx in Shenzhen.^e Most of the founding team members were veterans from banks or other FIs. Tang, named as Chairman, and Zeng Xuhui, named as President, both had extensive experience working as executives at multiple FIs, which included not only traditional FIs such as China Construction Bank but also emerging financial firms focusing on the microfinance market such as Zhong An Credit and Home Credit. **Their comprehensive experience in the financial industry gave the CredEx start-up team unique insights into the microlending industry.**

First, the team saw the huge demands and opportunities in the industry. In Shenzhen, the starting point of their new company, migrants comprised more than 90% of the population;⁴ entrepreneurship and innovation pervaded the air; and the thirst for capital surged as the number of SMEs grew exponentially. The CredEx team estimated that in 2010, the market demand for microfinance in Shenzhen alone would reach RMB 600 billion. Second, the team was deeply aware that the existing business and management models of the traditional FIs, such as **banks, could not convert the huge demand for microfinance into business opportunities that were profitable and scalable in a sustainable manner. The only way to capture the opportunities in the microfinance market was through innovating the business models and management models.** Third, their experience working in traditional FIs equipped the CredEx start-up team with a keen insight into the patterns of development of the credit business. One of these patterns was the **high correlation and balance between scale and risk.** Rapid expansion often resulted in higher default risk, but excessively tight control of default risk could in turn slow down the growth of assets or even cause decline. The key to breaking the constraint was for a company to **improve its risk management capability.** Therefore, from the very beginning, the CredEx start-up team prioritized developing its risk management capability as one of the most important core competencies. It also served as a starting point for CredEx to innovate and to transform the credit model of traditional banks.

The “Credit Factory” Model

At its founding, CredEx replaced the IPC model with the “Credit Factory” model in its loan approval process. The Credit Factory model differed from the IPC model mostly in **centralizing the authority of loan review and approval at the headquarters.** The front-end offline branches carried only two functions: customer acquisition and services, which were supervised by the headquarters’ Marketing Department and Customer Service Department, respectively. In order to improve the efficiency and quality of risk management, CredEx set up an independent review and approval center at its headquarters. Drawing on the management model for factory assembly lines from manufacturing industries, CredEx adopted a very refined division of labor and broke down the process of risk management into highly specialized jobs. For instance, the review part was divided into data review, phone review, and web review. The functions of credit line preassessment and final approval in the approval process were also separated. Such a specialized division of labor not only improved the

^e The company was originally founded as Shenzhen Zhongxing Microfinance Limited and later changed its name to Shenzhen CredEx Finance and Technology Co. Limited.

efficiency of each risk management step but also greatly increased the independence of each review and approval phase, thus reducing the moral hazard.

The innovative Credit Factory model laid a good foundation for CredEx to further informatize and optimize its loan approval model. CredEx invested heavily in technological R&D projects and digitally optimized each risk management context. First, CredEx applied a series of tech methods to ensure the authenticity of customer data and the efficiency of digitizing customer information. For example, CredEx developed a field inspection app targeting its business loan clientele, which standardized the mandatory on-site actions for information collection and documentation in field inspections. CredEx thereby significantly increased the proportion and level of standardization of objective information in the customer data, such as photos and videos, which were used as substitutes for the subjective judgment of the on-site investigators. Also, the app would send the customer data back to the company's headquarters to be saved in the Credit Factory at the earliest possible time, which greatly improved the speed of information access at the headquarters and its review and approval process.

Second, CredEx built its own information system to support its Credit Factory and further optimize the efficiency and effectiveness of the review and approval process. The information system randomly assigned orders to different steps throughout the process to ensure independent operation of each review and approval step. With the information system, CredEx could apply the full set of rules, without deviation, in the review and approval process, further reducing the impact of arbitrary, subjective judgment on determining credit results. The information system also enabled CredEx to dynamically deploy resources in response to the efficiency and status of each review and approval step, changing the whole risk management process from a linear operation to a parallel operation and optimizing the overall procedural efficiency.

While reducing the microlending risk, CredEx's Credit Factory also improved the customer experience, cutting the industry-average waiting time of three days between application submission and loan issuance to one day. The innovation of the Credit Factory model won CredEx the Financial Innovation Award issued by the Shenzhen Municipal Government.

Early Business Model Innovation

CredEx also initiated other business model innovations in its early days. One of those innovative measures was the choice of clientele. Although CredEx was able and qualified to provide loans of up to RMB 5 million and was fully aware of the higher profitability of these jumbo loans, it still defined its products as targeting loans below RMB 500,000. At this scale, the loans were mainly for personal consumption or working capital/expansion of micro, small, and medium-sized enterprises (MSMEs). CredEx designed two products, Salary Loan and Business Loan, to meet the respective needs of these two segments. The loan periods ranged from 12 to 24 months. Because it was difficult to obtain information on MSMEs, CredEx's early loan products were primarily Salary Loans made to consumers.

Another innovation had to do with the source of funding. Since microlenders were not qualified to take deposits like banks, most of them relied on their own capital for development and expansion. However, CredEx found that although the risk management model, cost structure, and culture did not provide a great foundation for traditional FIs to engage in the microlending business, the high returns on microloan-based assets were still somewhat attractive to these FIs. On the other hand, government regulatory commissions promoted and required the participation of traditional FIs in providing inclusive financing, which also created strong demand from traditional FIs such as banks for microloan-based assets.

Based on this insight, CredEx designed its **Assisted Loan Model**, leveraging the partner bank's **capital at a fixed ratio** (maximum 1:10) and helping the bank issue microloans below RMB 500,000. In the Assisted Loan Model, the bank was mainly responsible for providing the capital for loans while having the right to finally approve loans. CredEx oversaw all other steps prior to the final approval, such as acquiring customers; collecting, reviewing, and evaluating customer information; and preapproving customers. For return distribution, **the bank received a premium fixed loan interest return; CredEx got the remaining interest** return once the fixed return for the bank was deducted while solely assuming the customer acquisition cost, the loan default risk, and all related operational costs. The **Assisted Loan Model eliminated the capital bottleneck of CredEx's growth**. The company's annual loan balance spiked year by year, increasing from RMB 220 million in 2011 to RMB 1.32 billion in 2014. (See **Exhibit 7** for CredEx's financial and operational data from 2011 to 2017.) In 2013, CredEx had a total of 33 offline branches and nearly 2,000 employees, of which almost 1,300 were members of the sales force. (See **Exhibit 8** for changes in the numbers of CredEx branches and the company's HR structure from 2011 to 2017.)

CredEx: Journey of Digital Transformation (2014–2017)

The Microfinance Landscape from 2014

The year 2014 was a turning point for CredEx. Tang and his team sensed that the landscape of the microfinance industry was experiencing some subtle changes. The number of microlenders, which was less than 3,000 in 2010, tripled in 2014 to close to 9,000.⁵ The scale of loan balance also surged from about RMB 200 billion in 2010 to more than RMB 900 billion in 2014.⁶ That year also saw new business models emerging in the microfinance industry and competing against established microlenders for clients. **P2P platforms that deployed a point-to-point matching model to address microfinance demands experienced explosive growth**, tripling from 657 at the beginning of 2014 to 2,290 at the end.⁷ Consumer credit financing companies targeting big-ticket-item consumption (such as cell phone purchases) also entered the market in 2014. Heavyweight traditional FIs, including China Merchant Bank and China Industrial Bank, started partnerships with telecommunication companies such as China Unicom to provide consumption loans and increased their investment in the microfinance market. Two internet giants with the biggest distribution and data advantages also entered the microfinance business separately: **Alibaba started Ant Financial in October 2014 and launched its microfinance product, Ant Credit Pay, later that year; and Tencent jointly founded WeBank in December 2014** with mostly a couple of local companies in Shenzhen. Utilizing social and payment data accumulated on WeChat, the microfinance product Weilidai entered the market the following year, with a focus on the low-end segment. Suddenly, the microfinance industry became unusually "packed." (See **Exhibit 3** for detailed information on the growth of the P2P and internet finance industry from 2014 to 2017.)

The sudden influx of competitors eventually resulted in vicious competition. In December 2014, salespeople from the two big microlending companies got into a knife fight while competing for customers, leaving one death and two injuries.⁸ The competition also caused a rapid increase of customers' debt ratio, and risk was aggregated at an accelerated manner. One driver of the risk aggregation was the shift of bargaining power toward the industry's salespeople. Specifically, salespeople were shopping their customer resources to several microlenders, which directly resulted in overleveraged customers. More capable salespeople even started to mediate between microlenders and other salespeople, peddling customer information for fees and pushing up microlenders' customer acquisition costs.

Offline-to-Online (O2O) Model

As a leading company in the industry, CredEx had not been greatly affected by the competition in 2014. Its revenue increased that year by more than 50% to nearly RMB 630 million, up from RMB 400 million in 2013. The nonperforming loan (NPL) ratio rose from 1.83% to 2.36%, but it was still significantly lower than the industry average. However, the series of changes in the industry led Tang and his team into deep reflection. In response to the chaos happening at the sales end of the microlending business and the shift of bargaining power, CredEx developed an offline-to-online (O2O) model, launching its Fei Xia app in April 2014. Upon registration on the Fei Xia app, any salesperson in the industry could immediately become part of CredEx's customer acquisition network and recommend customers for CredEx using Fei Xia.

While launching the app, CredEx also started restructuring the functions of its offline branches, shifting their focus from sales to providing services for customers and salespeople. At the same time, CredEx gradually changed the status of its salespeople at the branches from employees to independent contractors and restructured its risk management process accordingly. The initiative drastically reduced the labor cost at the sales end while doubling the number of customer loan applications. From 2013 to 2014, the average annual loan balance managed by CredEx increased from RMB 800 million to RMB 1.3 billion.

But the success in this phase did not relieve the anxiety of Tang and his team about the future of the company. Tang realized that competition and imitation were quickly changing the microfinance industry from a blue ocean, where there was little competition, to a red ocean. More important, although the CredEx start-up team was known for leading innovation in the industry, it became more and more difficult for them to find new ways to innovate their offline model, which was based on heavy operations. Competition in the microfinance industry resulted in more and more homogeneous product and service offerings. Compared with microlenders backed by banks or internet companies, independent microlenders suffered from an inherent disadvantage in capital costs and existing data resources. How could CredEx keep innovating its business model to build a sustainable competitive advantage? This question challenged Tang and his team.

Customer Analysis

In his quest for answers, Tang and his team engaged the entire company to dive into the market, investigating customer needs and looking for solutions. Each executive on the team conducted 15 to 20 surveys during that period, finding friends who were small business owners or big spenders to understand the users' pain points in accessing and using microfinance products. Based on the analysis of hundreds of interviews and questionnaires, the CredEx team identified four characteristics of microfinance loans: "short term," "small amount," "frequent uses," and "high urgency." Specifically, "short term" meant the loan was needed for only a short time period, typically a few months or even shorter; "small amount" meant the amount was typically as small as RMB 30,000 to 50,000; "frequent uses" meant such funds were frequently and irregularly needed due to their nature of emergent backup and supplementary funding; "high urgency" meant such funds were often needed for unexpected reasons and so were usually needed urgently. Also, due to their much higher interest rates (compared to those of bank loans), microfinance loans were mainly used for working capital or bridge financing. The borrowers tended to reduce the amount and period of the loan as much as possible so they could lower costs. At the time, the microfinance products available in the market could not fully address these needs.

The CredEx team also found five major pain points of microfinance customers: (1) difficulty applying, (2) difficulty obtaining approval, (3) difficulty using the money, (4) difficulty repaying, and (5) difficulty borrowing again. At the time, a customer had to collect and prepare a large amount of paper files such as bank statements, visit the offline branch at a specified time to submit the application, and wait for a few days or even more than a week to get the results of the loan review. Microfinance products sometimes also imposed complex and demanding terms, such as requiring customers to deposit a certain percentage of the borrowed funds in a designated account or purchase wealth management products with a portion of the funds. In addition, these products often had stringent loan maturity terms, prohibiting the borrowers from paying off the loan prior to the maturity date. **If an early repayment was requested, the borrower often had to pay a penalty to the lender. After repayment, if a borrower wanted to apply for another loan, the cumbersome process had to start again.**

These findings brought Tang and his team both excitement and perplexity. They were excited because the pain points suggested the direction for innovation. The CredEx team believed that developing a product to address these pain points would differentiate CredEx from its competitors. The team was perplexed because the O2O model (e.g., Fei Xia) still relied on heavy operations, leaving very limited room for continuous optimization or scale-up. They did not have a clear vision on how the new business model should be designed. It was just a vague intuition that the next business model innovation would inevitably involve the shift from offline to online.

Formulating a Strategy for the Age of the Mobile Internet

The technology infrastructure of business also went through dramatic change in 2014. According to Andreessen Horowitz, a renowned U.S. venture capital firm, shipments of smartphones with iOS or Android systems dwarfed shipments of personal computers in 2014.⁹ Smartphones became the most important end-user device around the world. Annual smartphone shipments in China also increased year by year, reaching 400 million units for the first time in 2014, an amount almost double that of units shipped in 2012. The number of mobile internet users in China reached 550 million in 2014, up from 420 million in 2012 and a 150% increase from 2010, the year when CredEx was founded. (See **Exhibit 4** for details of development of mobile internet and smartphones in China.) The number of mobile apps for smartphones increased from around 900,000 in 2012 to more than 3 million in 2014.¹⁰ In comparison, this number was a little more than 150,000 in 2010.¹¹ At the end of 2013, China's Ministry of Industry and Information Technology started issuing TD-LTE licenses to telecom operators. China Mobile became the first operator in China to deploy 4G networks. With 4G, data transmission speed spiked from between 144 kilobits per second (Kbps) and 2 megabits per second (Mbps) in the 3G era to between 100 Mbps and 1 gigabit per second (Gbps), greatly expanding the possible application scenarios of mobile internet. Technological developments also brought far-reaching changes to user behaviors. For example, users gradually developed the habit of interacting through the mobile internet. In 2014, the smartphone-based social media app WeChat had about 500 million monthly active users.¹² Many social and consumer behaviors in daily life raced to transfer to the mobile internet.

Although the external environment was ripe for online operation, the CredEx team still had many open questions about bringing its business online. The concerns mainly came from two aspects: (1) the debate about the way to go online, such as whether CredEx should give up its operational advantages accumulated offline and completely go online, and (2) whether CredEx had the resources and capabilities to go online. In 2014, although smartphones were already loaded with apps, very few of them targeted microfinance demands.

At the beginning of 2015, Tang took his start-up team to Yunnan province. It was billed as a rapeseed blossom trip (named for the bright yellow flowers of *Brassica napus* that grew in the area), but it was

actually aimed at helping the team reach a consensus on the company's future direction via closed-door meetings and brainstorming sessions. After a few days of heated discussions and clashes of ideas, the team finally agreed that CredEx should completely discard the previous business model of heavy offline operations and provide its customers with a completely online microfinance app based on mobile internet technology.

The product was named after the company, "CredEx." Targeting the pain points of the offline customer experience, the CredEx team designed the app's positioning and features as "borrow-and-repay-as-you-go, anytime and anywhere." "Anytime and anywhere" aimed to use the smartphone device to make the customer's loan application more convenient. "Borrow-and-repay-as-you-go" targeted the four characteristics "short term, small amount, frequent uses, and high urgency" associated with microfinance loans, which led to the redesign of CredEx's microloan product to a microcredit product in order to tackle those problems of using the funds and making the repayment. The positioning sounded simple and clear; however, its execution faced complicated challenges and huge difficulties. The first and foremost challenge was risk management. To realize "anytime and anywhere," CredEx needed to make quick decisions and provide feedback in a short period (e.g., in a few minutes) after the users submitted their applications. With a manual review and approval process, it would take CredEx at least 40 minutes to process a single application. Clearly, CredEx needed to build a brand-new risk management system, using computers, data, and algorithms to replace humans in decision-making.

Building a New Risk Management System

At the beginning of 2015, CredEx formally started to explore the digital risk management system. The first step was to lay out all the nodes and modules in the risk management process with high granularity and specify the resource demands for each step. Data was the most important raw material for a digital risk management system. However, at the time, standardized and systematic data resources were still scarce. Luckily, as a member of the Microfinance Company Pilot Programme organized by the People's Bank of China (the nation's central bank), one of CredEx's subsidiaries was granted access to the central bank's personal credit data. This valuable data source also became the starting point for CredEx to perform digitalized risk management. Drawing on experience from the leading international credit rating agencies, CredEx built its quantitative risk management system in about six months, mainly relying on personal credit data to make risk judgments. This system set up a good foundation for allowing CredEx to quickly debut its first product.

At the same time, CredEx poured resources into marketing to promote its app, flooding its targeted customer segments with billboard, radio, and online ads. Based on the user psychological positioning theory, CredEx segmented the market and defined its product as "the First Cellphone Loan App in the World," emphasizing the app's quick and convenient features, such as "determine your credit limit in five minutes and four steps." The targeted publicity quickly pushed CredEx's microfinance app to the forefront of users' visibility and reinforced its uniqueness. The launch of the CredEx app showed that the investment in branding was very effective, but such investment also directly led CredEx to a net loss of RMB 68 million in 2015, plummeting from a net profit of RMB 68 million in 2014.

CredEx 1.0

In October 2015, CredEx held a product launch event in Beijing, announcing that the app CredEx 1.0 had gone online and that external testing had started in Shenzhen. However, the product ran into many unforeseen problems as soon as it went online. First, the launch of the app led to a sharp increase of credit applications. In 2014, prior to the launch, CredEx had reached a peak of 1,680 average monthly

applications in Shenzhen. When CredEx 1.0 went online, the average *daily* application volume in Shenzhen suddenly reached 1,700, which directly paralyzed the 100-megabyte bandwidth server in the CredEx back office. Second, CredEx 1.0 encountered many malfunctions—which included stutters, flashes, crashes, and in some cases an interface size that exceeded the screen size—resulting in an extremely **disappointing user interface experience**. Third, as the dimensions of its risk management data were relatively limited, CredEx 1.0 had a very low loan approval rate in order to avoid risk; the mobile application process itself had multiple unreasonable and unclear spots. These factors caused a very high user drop-off rate in the application process. The overall credit experience needed abrupt and drastic improvement.

CredEx immediately started improving and optimizing its app after the launch. To improve the customer experience, CredEx ranked all customer complaints by frequency and solved them in that order. At the same time, CredEx used the customer behavior data recorded by the CredEx app to figure out how to optimize the product. For example, by observing customers' interactions with the app in their loan application process, CredEx found that customers most often dropped off when being asked to add their bank cards. Specifically, the original application process required users to add the bank card they would be using for repayment before CredEx gave them their credit line information. Based on this observation, CredEx swiftly switched the **add-a-card step to come after the credit line information so that users would have to provide their bank card information only when they needed to access the funds**. This adjustment greatly lowered the customer drop-off rate in the application process. CredEx also kept simplifying and optimizing the application process, reducing the initial **15 user-filled information fields to 9 fields**. Some complicated information, such as user mailing address, previously needed to be manually filled in. CredEx simplified the required information and changed the menu to a click-and-choose format.

Behind these customer experience improvements was the drastic change that CredEx made in its digital risk management model. In the review stage, the offline risk management model focused on moral hazards, while the online model emphasized third-party fraud prevention. The rise of online microcredit business gave hackers and other tech-savvy lawbreakers the opportunity to make a profit by exploiting the technical loopholes of the microlenders. Since there was no salesperson verifying user identity in person, the online review process needed to use data (such as the user's ID card number, phone number, and behavioral analytics), technical methods (such as device fingerprints and face recognition), and risk control rules to remotely verify the authenticity of the user's loan application. For example, the device fingerprint technology could obtain the device ID unique to each smartphone and apply back-end risk control rules to identify potential fraudulent scenarios such as multiple customers applying with one device ID.

Another important antifraud approach was the cross-validation and filtering of multisource data. Since the beginning of its transformation to the online model, CredEx had started honing its ability to obtain, process, and apply data, actively investing in and exploring the field of big data. Instead of relying only on personal credit data from China's central bank, CredEx contacted hundreds of domestic data providers and tested their data; this allowed the company to expand the number of its data sources to 20–30. These sources included verification data from the Ministry of Public Security, telecommunication firms, and UnionPay, as well as negative data such as blacklists, default records from CredEx itself, and other external data sources. **Through basing its loan reviews on multiple dimensions, CredEx could more accurately exclude applicants with high fraud risk** and reduce the probability of falsely rejecting applicants with low fraud risk. The enriched data resulted in a decline of CredEx's rejection rate at the antifraud stage and a significant spike of its approval rate without increasing its risk exposure. The customer loan application experience was improved as well.

CredEx also used big data in modeling its risk management strategy and credit line approval. When the CredEx decision-making engine went online in 2015, it had only around 100 rules. In less than three years, the engine accumulated nearly 1,000 risk management rules. Big data provided a vast number of data dimensions. As a data source, a credit report with complete and rich information about a single customer could potentially provide thousands of tags about that customer. It was completely out of the range for the human brain to process such massive amounts of information. CredEx used these tags to establish a variety of risk management rules and customer credit scorecards. Furthermore, big data made the credit line approval process more accurate and replicable. The offline model of the credit line review and approval system depended mainly on the reviewer's experience and the collective wisdom of loan review committee experts, which resulted in relatively low accuracy and poor replicability. For instance, CredEx did some internal experiments by sending some already approved loan applications back to their original reviewers. The deviation in the approved loan size exceeded 30%. With the big-data model, unless the data or the model changed, the final loan amount was always consistent and could be replicated. At the same time, the historical data could be applied to immediately predict and verify the effect of each adjustment to the model, which further improved the speed and accuracy of decision-making.

Data-driven risk management also led to a significant improvement in the efficiency of the review and approval process. With the offline model, it took about 200 to 300 CredEx employees to review and approve approximately 1,000 applications per day. Once the company switched to the online model, the risk management team dropped to about 80 people, yet those people completed tens of thousands of applications per day during the peak season. At the same time, the data-driven operation allowed CredEx to adjust and innovate its risk management strategies more dynamically and accurately. In 2017, CredEx reviewed and adjusted its established risk management strategies on a monthly basis. A new risk strategy could go online only after its potential effectiveness was verified by historical data. The efficiency and the accuracy of risk management decision-making were both significantly improved.

The launch of CredEx 1.0 led the company to discover the limitations of its underlying technological infrastructure such as its computing power and storage capacity. That discovery provided it with the motivation and the opportunity to further enhance its technological capabilities. The launch of 1.0 brought the CredEx tech team firsthand experience in dealing with the heavy traffic and high volatility of an online operation. The team was prompted to rethink the storage capacity, speed, and scalability of its existing underlying technological infrastructure.

As the first response, the CredEx tech team built its Supercomputing Mobile Technology Platform, which used middleware to solve the problem of data accessing and computing speed, and greatly improved the throughput, or transactions per second (TPS), and the maximum number of concurrent tasks of CredEx's technology system. Second, the CredEx tech team adopted distributed storage and data processing technologies while exploring other open-source technologies such as cloud container. The approach kept the cost of the underlying computing and storage capacity under control while continuously improving the system's speed and scalability. Traditional financial architecture mainly relied on partnership with the server or storage service providers to upgrade the underlying technology, which was not suitable for open-source technology. Open-source technology's rapid upgrades and iterations required the CredEx technology squad to constantly learn and develop its own technical capabilities so it could follow and use the open-source technology to optimize its own underlying technical architecture. CredEx therefore continued to invest in its tech team. Its annual R&D expenditure also kept rising, which in 2017 more than tripled compared with 2014.

CredEx 2.0 and 3.0

The rapid development in its product design, risk management, and technological infrastructure enabled CredEx to quickly iterate on its app products. The CredEx 2.0 version was released in February 2016, just **four months after the launch of CredEx 1.0**. The 2.0 version resulted in substantial improvements in customer interaction and experience, as well as the stability of its risk management model, compared to the 1.0 version. Moreover, CredEx 2.0 expanded its loan application regions from Shenzhen to 50 cities across the nation. In August 2016, CredEx launched the 3.0 version, which further optimized the risk management process and reduced the lead time between users submitting their application and receiving their credit line from five minutes to three minutes. “Change the world in three minutes” became CredEx’s new tag line to promote its app. Meanwhile, CredEx leveraged its continuously growing data capability to vigorously expand its client base to both higher-end and long-tail customers. On the one hand, that expansion enabled CredEx to make its microfinance products more inclusive. On the other hand, it enhanced the profitability of its asset portfolios.

The digitization of its business model led to a substantial adjustment of the CredEx organization and personnel structures. In particular, it resulted in a radical reduction of the company’s sales, customer service, and loan review labor force. Instead, the company hired more technical and product development employees. As a result of digitization, the total number of **CredEx employees dropped from over 2,000 to fewer than 400 in 2017**. At the beginning of 2015, after deciding to switch to the online model, CredEx restructured its organization. The original departments at the headquarters were replaced by six committees: the Strategy Committee, the Marketing Committee, the Product Committee, the Risk Control Committee, the Technology Committee, and the Operations Guarantee Center. These six committees were organized and coordinated by six senior executives at CredEx. Employees could apply to join the corresponding committees based on their own interests and capabilities, while the committee head would make the final decision based on the employee’s capability, action plan, and potential contributions. The CredEx senior executive team regularly participated in the work and discussions of a number of different committees; however, the head of each committee had the authority on decisions. **This organizational change broke the silos** of the original departments and promoted cross-functional cooperation. It also better motivated the executive team and the employees to fully use their capabilities.

The adoption of the online model also resulted in changes to the organization culture at CredEx. The biggest change was the establishment of the digital culture. In 2016, CredEx invested substantial resources to build its big-data platform, aiming to systematically manage and use the large volume of data it generated in its operation and ingested from external third parties. Furthermore, CredEx wanted to connect the data silos of various departments. The big-data platform first collated, labeled, classified, and stored the data. It then generated a series of standardized reports, known as the Data Mart, tailored to the needs of business units. Meanwhile, the platform integrated open-source tools for data extraction, analysis, and presentation, such as Python, R, and Tableau, to help internal business units better use data to make decisions.

The big-data platform greatly improved the efficiency of data collection for the purpose of decision-making. For example, it once took about 10 days for CredEx’s finance department to get the data needed for compiling financial statements (e.g., fetching the numbers and generating customized statements). After the platform standardized the templates of various statements, it only took a few seconds to get the required data. More important, the **big-data platform became the cornerstone to build up the digital culture at CredEx**. At the core of the data culture was data-driven business decisions. In other words, any discussion on business decisions—such as those on risk management strategies, customer positioning, and marketing—had to be based on the results and forecasts

suggested by data. This decision-making approach not only made the process more rigorous but also significantly reduced the cost of trial and error.

The digital transformation led to huge changes in CredEx's scale of business and cost structure. From 2014 to 2017, the average annual loan balance of CredEx rose from RMB 1.3 billion to RMB 4.25 billion; the number of loan customers rose from 43,000 to 97,000; revenue jumped from RMB 600 million to RMB 1.4 billion; and net profit rose sharply from RMB 68 million to RMB 500 million. However, such an expansion did not lead to higher risk: the NPL rate only changed from 2.36% in 2014 to 3.06% in 2017. At the same time, CredEx cut its labor cost as a percentage of total cost from 30%–40% during the offline era to 5% in 2017, while the percentage of its R&D expenditure rose from below 5% before 2015 to 12% in 2017.

2017: Changes in the Institutional and the Technological Environment

In 2017, the microfinance industry came to a crossroads between institutional and technological change. The year 2016 saw a large-scale default in P2P platforms as well as a number of vicious incidents resulting from student loans. Both brought widespread negative social impact and resulted in the shift of the regulatory attitude toward online lending from encouragement and support to stringent supervision. Since the second quarter of 2016, financial regulatory authorities had announced a series of specific operational requirements for the industry. In 2017, the government agencies intensively issued various rectification requirements for the microlending industry, such as scaling back businesses, suspending license issuance, requiring record-filing and registration of the P2P industry, and performing on-site inspection. The high pressure from the regulatory bodies peaked at the end of 2017 with the issuance of Circular 141, restricting the Assisted Loan Model with regard to risk management, guarantor qualifications, leverage ratios, and other aspects. More important, the shift in regulatory attitude made traditional financial institutions extremely cautious about working with microlending companies, which in turn led to a cutoff of capital as well as spiking capital costs for microlenders. Many microlenders were forced to downsize or even close their businesses. The industry expected this regulatory pressure to become the new norm. (For details of the policy changes of the online lending industry imposed by China's regulatory authorities, see **Exhibit 6**.)

In 2017, China's microlenders faced not only the challenge of increased regulatory pressure but also increased competition and fewer qualified clients. (See **Exhibit 5** for details on new entrants to the online finance and consumer credit industry.) Internet microfinance had developed rapidly in the past few years. For instance, WeBank's Weilidai cumulatively issued RMB 870 billion in microloans after its launch only three years before.¹³ Ant Financial's counterpart product, JieBei, also reached RMB 300 billion in scale.¹⁴ The loan balances or transaction volumes of several other microfinance companies also reached tens of billions of RMB. Like CredEx, many of these companies used an online model for customer acquisition and loan issuance, resulting in a sharp rise in online customer acquisition costs. Specifically, such costs more than tripled. On the other hand, the leverage ratio of customers increased significantly. Customers increasingly borrowed from multiple lenders, which led to a sharp increase of the "joint debt" risk. Loan demand and loan default rates were both rising rapidly.

Unlike the regulatory environment and the competitive environment, however, the technological environment of the microlending industry was still evolving in a positive direction. In 2017, the National Development and Reform Commission issued a notice calling for pilot projects to scale up the 5G network in 2018, which signaled the speed-up of the construction of the 5G network. Once implemented, the 5G network technology would increase the data transmission speed from the 4G era's 100 Mbps–1 Gbps to about 20 Gbps. At the same time, the rapid development of mobile internet paved the way for various industries to significantly improve their digitization levels and brought

abundant data resources. Many new data resources, such as GPS data and mobile phone battery data, had been tentatively deployed in financial scenarios. In technology infrastructure, container technology had developed and matured rapidly, and was increasingly used in real operations. Container technology was a series of software-as-a-service (SaaS) and platform-as-a-service (PaaS) products that used virtualized operating systems to develop and deliver packaged software. With features such as low cost, easy scalability, and open-source software, it provided enterprises with cloud computing platform-based tools for rapid and agile innovations.

In 2017, under siege from the internet companies, traditional FIs began to pay more attention to online operations and fintech innovation. China Merchants Bank (CMB) and Ping An Bank, respectively, proposed the new positioning as “Fintech Bank.” CMB also decided that it would continuously set aside 1% of the previous year’s pretax profit, which was RMB 790 million in 2017, to set up a fintech fund for the incubation and funding of internal projects, which drove CMB’s fintech transformation.¹⁵

The convergence of institutional changes and technological changes presented a scenario that fit Dickens’s famous opening words in *A Tale of Two Cities*: “It was the best of times, it was the worst of times.” The new challenge faced by CredEx could be summed up in the following question: As a leader of business model and technological innovation in the microlending industry, how should CredEx continue to innovate its business model in such an environment to better connect market demand with its core competence and build a lasting competitive advantage in the digital age?

Exhibit 1 China's Economic Development, 2007–2017

	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
GDP Growth Rate (%)	14.2	9.7	9.4	10.6	9.6	7.9	7.8	7.3	6.9	6.7	6.8
Per Capita Disposable Income of Urban Residents (RMB)	13,603	15,549	16,901	18,779	21,427	24,127	26,467	28,844	31,195	33,616	36,396
Final Consumption Expenditure (trillion)	13.6	15.7	17.3	19.9	24.1	27.1	30.0	32.8	36.2	40.0	43.7
Loans for Consumption Use from All Financial Institutions (trillions)	3.3	3.7	5.5	7.5	8.9	10.4	13.0	15.4	19.0	25.0	31.5
Medium and Long-term (Trillions)	3.0	3.3	4.9	6.5	7.5	8.5	10.3	12.1	14.9	20.1	24.7
Short-term (Trillions)	0.3	0.4	0.6	1.0	1.4	1.9	2.7	3.2	4.1	4.9	6.8

Source: Created by casewriter using information from National Bureau of Statistics of China, <http://www.stats.gov.cn/english/>, accessed January 2020. .

Exhibit 2 The Growth of the Microfinance Industry in China, 2009–2017

	2010	2011	2012	2013	2014	2015	2016	2017
Number of Microlenders	2,614	4,282	6,080	7,839	8,791	8,910	8,673	8,551
YoY Growth (%)	96	64	42	29	12	1	-3	-1
Number of Employees	27,884	47,088	70,343	95,136	109,948	117,344	108,881	103,988
YoY Growth (%)	N/A	69	49	35	16	7	-7	-4
Loan Balance ('100 Million RMB)	1,975	3,915	5,921	8,191	9,420	9,412	9,273	9,799
YoY Growth (%)	20	98	51	38	15	0	-1	6

Source: Created by casewriter using information from The People's Bank of China, <http://www.pbc.gov.cn/diaochatongjisi/116219/116319/index.html>, accessed January 2020. .

Exhibit 3 China's P2P Industry and Internet Consumer Finance Development, 2014–2017

	2014	2015	2016	2017
P2P Monthly New Transaction Volume ('100 million RMB)	371	1,337	2,443	2,248
YoY Growth %	215	261	83	-8
P2P Loan Balance ('100 million RMB)	1,036	3,882	6,795	10,418
YoY Growth %	236	275	75	53
P2P Platforms	2,290	3,543	2,670	2,351
YoY Growth %	249	55	-25	-12
Internet Consumer Finance Loans Issued ('100 million RMB)	183	1,184	4,367	43,847
YoY Growth %	205	546	269	904

Source: Compiled by casewriters using information from: www.wdzj.com, iResearch, and WIND, accessed January 2020.

Exhibit 4 China's Mobile Internet and Smart Phone Development, 2011–2017

	2011	2012	2013	2014	2015	2016	2017
Smart Phone Shipments ('100 million)	1.56	2.13	3.51	4.23	4.30	4.67	4.44
YoY Growth %	36.5	64.8	20.5	1.7	8.6	-4.9	36.5
Mobile Phone Internet Users ('10,000)	35,558	41,997	50,006	55,678	61,981	69,531	75,265
YoY Growth %	18.1	19.1	11.3	11.3	12.2	8.2	18.1
Percentage of Mobile Phone Internet Users in the Total Internet Users	69.3	74.5	81.0	85.8	90.1	95.1	97.5
YoY Growth %	7.5	8.7	5.9	5.0	5.5	2.5	7.5
Mobile Internet Data Usage ('100 million GB)	5.4	8.6	12.7	20.6	41.9	93.8	246
YoY Growth %		59.26	47.67	62.20	103.40	123.87	162.26

Source: Compiled by casewriters using information from the China Internet Network Information Center (CNNIC), and International Data Corporation (IDC).

Exhibit 5 New Competitors of Internet Finance and Consumer Credit

	Founded	Main Products	Strategic Shareholders	Business models	Development (2017)
Ant Financial	Oct 2014	Spending installments, cash loan	Alibaba	Internet finance holding	Ant Credit Pay (credit installment product) Revenue 2.99 billion Net profit 2.04 billion JieBei (Cash loan) Revenue 9.26 billion Net profit 5.99 billion
WeBank	Dec 2014	Cash loan	Tencent	Internet banking	Registered users >60 million Loan balance 47.7 billion Revenue 6.75 billion Net profit 1.45 billion
Pingan Puhui ^a	2015	Cash loan	Ping An Insurance	Internet microloan	Users 5 million Accumulated loans issued: 400 billion RMB (as of August 2017) ^b
Merchants Union Consumer Finance Qudian	Aug 2014	Spending installments	China Merchants Bank, China Unicom	Offline consumer finance	Total asset 46.98 billion Revenue 4.16 billion Net profit 1.19 billion
	Mar 2014	Spending instalments, cash loan	Ant Financial	P2P + Assisted loan	Registered users with credit approval 26.2 million Loan balance 11.2 billion Revenue 4.78 billion Net profit 2.23 billion
Lexin	Oct 2013	Spending installments	JD	Online consumer finance	Registered users >20 million Loan balance 19.27 billion Revenue 5.9 billion Net profit 240 million

Source: Annual reports; and China Merchants Securities, "Contributing profits in 2018 exceeded 10 billion. Spending money and borrowing money is actually a very profitable business!" Sohu.com, January 12, 2019, http://www.sohu.com/a/288451603_436214, accessed December 2019.

^a Founded in 2005, Ping An Puhui Investment Consulting Co., Ltd. started its personal consumer finance business in Shenzhen. In 2015, Ping An credit insurance business, Ping An direct loan Business, Lufax's P2P microloan business, and the Fullerton financing guarantee business were integrated into the Ping An Puhui Financial Business Group, focusing on addressing the consumer financial needs of individuals as well as small and micro enterprises.

^b "Ping An Puhui: 3 products lines prop up the consumer gold business, cumulative lending of RMB 400 billion," P2P Eye.com, <https://news.p2peye.com/article-507435-1.html>, accessed December 2019.

Exhibit 6 Policy Changes of the Microlending Industry Imposed by China's Regulatory Authorities

Date of publication	Document Title	Regulatory Authorities	Main Content
November 2013	Decision of the CCCPC on Some Major Issues Concerning Comprehensively Deepening the Reform	The Third Plenary Session of the 18th CPC Central Committee	Develop inclusive finance, encourage financial innovations, and enrich the financial market with more levels and more products.
April 2014	CBRC Press Release	China Banking Regulatory Commission (CBRC)	Specify the four bottom lines of P2P: all platforms are specified as intermediaries; no platform can act as guarantors; funding pools are prohibited; and illegally acquiring funds from the public is prohibited.
January 2015	Set up the Financial Inclusion Affairs Department	CBRC	P2P online loans are included in Inclusive Finance, meaning the "Inclusive Finance" nature of the P2P industry has been recognized by the regulators.
July 2015	The Guiding Opinions on Promoting the Healthy Development of Internet Finance	People's Bank of China and 9 other regulatory agencies	Specify the definition of Internet Finance for the first time. Specify the corresponding regulators for various categories of Internet finance business. According to the overall requirement of "encourage innovation, prevent risk, maximize the benefits and minimize the bad, and promote healthy development," set up a series of policies and measures that encourage innovation and support the stable development of internet finance.
March 2016	Guiding Opinions on Increasing Financial Support for New Consumption Areas	PBOC, CBRC	Give great impetus to the consumer finance market, actively build the organizational system of consumer finance, and continuously promote innovation in management modes and products of consumer credit.
October 2016	Plan to Implement Special Rectification Work on Internet Finance Risks	General Office of the State Council	More than 10 regulatory agencies including PBOC-issued regulations to implement special rectification of multiple categories including cross-sector finance business, third-party payment, P2P online lending, equity crowdfunding, and internet insurance.
November 2016	Guidelines for the Administration of Recordation and Registration of Online Lending Information Intermediary Institutions	CBRC, Ministry of Industry and Information Technology, and State Administration for Industry and Commerce	Provide specific guidelines on proper recordation of online lending intermediary institutions.

Exhibit 6 (continued)

Date of publication	Document Title	Regulatory Authorities	Main Content
April 2017	Guiding Opinions on Risk Prevention and Control of the Banking Sector	CBRC	During operation, online lending platforms must ensure that the lenders' funds are from legitimate sources; fraudulent, fake advertising is prohibited; strictly executive relevant regulations on interest rates in private loans are issued by the Supreme People's Court; illegal activities such as usury and debt collection through violence are prohibited.
May 2017	Notice on Further Strengthening the Regulation and Management of Campus Credit	CBRC, Ministry of Education, and the Ministry of Human Resources and Social Security	Online lending institutions must suspend all campus loan businesses to eradicate activities such as usury and debt collection through violence that infringe on the safety of college students.
August 2017	Notice on Implementing the Requirements for the Next Phase of Clean-up and Rectification	Office of the Leading Group for the Special Campaign against Internet Financial Risks	Business cannot scale up. Existing noncompliant business must be reduced. Additional noncompliant business is prohibited.
November 2017	Notice Concerning the Immediate Suspension of the Approval of Internet Micro-lending Companies	Office of the Leading Group for the Special Campaign against Internet Financial Risks	Immediately suspend approval of online microlending licenses.
December 2017	Notice on Regulating and Rectifying the "Cash Loan" Business	Office of the Leading Group for the Special Campaign against Internet Financial Risks, Office of the Leading Group for the Special Campaign against Peer-to-peer Lending Risks	Specify comprehensive regulation, clean up and rectify online microloans: strictly regulate and administrate online microlending business, suspend issuing online microloans that have no specific consuming scenarios and spending purposes, gradually reduce existing business, complete the rectification within the set deadline.
March 2018	Internet Finance Late Payment Collection Self-Discipline Pledge (for Trial Implementation)	National Internet Finance Association of China (NIFA)	Regulate internet finance late-payment collection behaviors. Practitioners shall be held liable if the legal rights and interests of creditor, debtor, and other parties involved are infringed due to poor outsourcing management.
August 2018	Notice on Reporting Debt Evasion Information of Borrowers on P2P Platforms	Office of the Leading Group for the Special Campaign against Internet Financial Risks	In order to crack down on malicious debt evasions of borrowers, related information should be reported as soon as possible.

Source: Compiled by casewriters from; the People's Bank of China, China Banking Regulatory Commission, the Supreme People's Court, and the National Internet Finance Association of China.

Exhibit 7 CredEx Finance and Operation Data 2011–2017

	2011	2012	2013	2014	2015	2016	2017
Loan Balance Managed	403.62	494.52	1114.52	1522.24	1056.12	5,328.45	3,716.66
Average Balance	215.74	449.07	804.52	1318.38	1278.73	3,510.17	4,256.61
Accumulated Credit Issued	-	-	-	-	51.28	20,807.84	27,304.36
Total Revenue	116.54	208.38	401.24	625.79	584.57	1271.91	1414.15
Interest Income	112.33	207.14	265.03	269.66	208.74	195.84	311.29
Non-Interest Income	4.21	1.24	136.22	356.13	375.83	1076.07	1102.86
Total Expenditure	107.54	170.04	332.69	553.8	679.6	1104.42	787.83
Interest Expense	16.08	30.65	62.96	100.18	90.26	159.26	207.18
Labor Cost	44.22	57.78	115.41	95.45	98.89	59.67	38.98
R&D Expense	-	3.51	5.1	30.87	40.28	72.14	97.89
Net Profit	6.83	28.88	50.02	68.18	-68.29	240.86	541.02
NPL %	1.69	1.39	1.83	2.36	8.20	3.22	3.06
No. of Customers that Withdraw Fund	13,267	12,554	35,873	43,272	21,884	148,919	97,253
No. of Fund Withdrawal Transactions	13,267	12,554	35,873	43,272	22,306	410,711	325,507

Source: Company documents.

Exhibit 8 CredEx Employees and Branches Data

	2011	2012	2013	2014	2015	2016	2017
No. of Offline Branches	13	19	33	36	30	0	0
Employees							
Sales/Marketing	338	510	1298	480	282	144	63
Headquarters	13	83	82	193	118	144	63
Branches	325	427	1216	287	164	0	0
Customer Services	103	107	192	199	129	0	0
Headquarter	8	3	22	31	20	0	0
Branches	95	104	170	168	109	0	0
Risk Control	92	137	312	419	308	158	120
Review and Approve	63	90	235	305	188	11	5
Strategy/Model	0	0	0	0	4	7	15
Collection	29	47	77	114	114	75	73
Technology	9	29	42	64	51	60	79
System construction	0	7	8	8	10	23	21
Application development	7	17	26	48	33	30	50
Others	2	5	8	8	8	7	8
Big Data	0	0	0	0	0	3	8
Products	0	0	0	0	0	0	5
Management	40	71	73	110	103	78	82
Total	582	854	1917	1272	873	443	357

Source: Company documents.

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